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Project #010: Influencing Outcomes through an Evidence-based Nurse-Driven Telemetry Discontinuation Protocol

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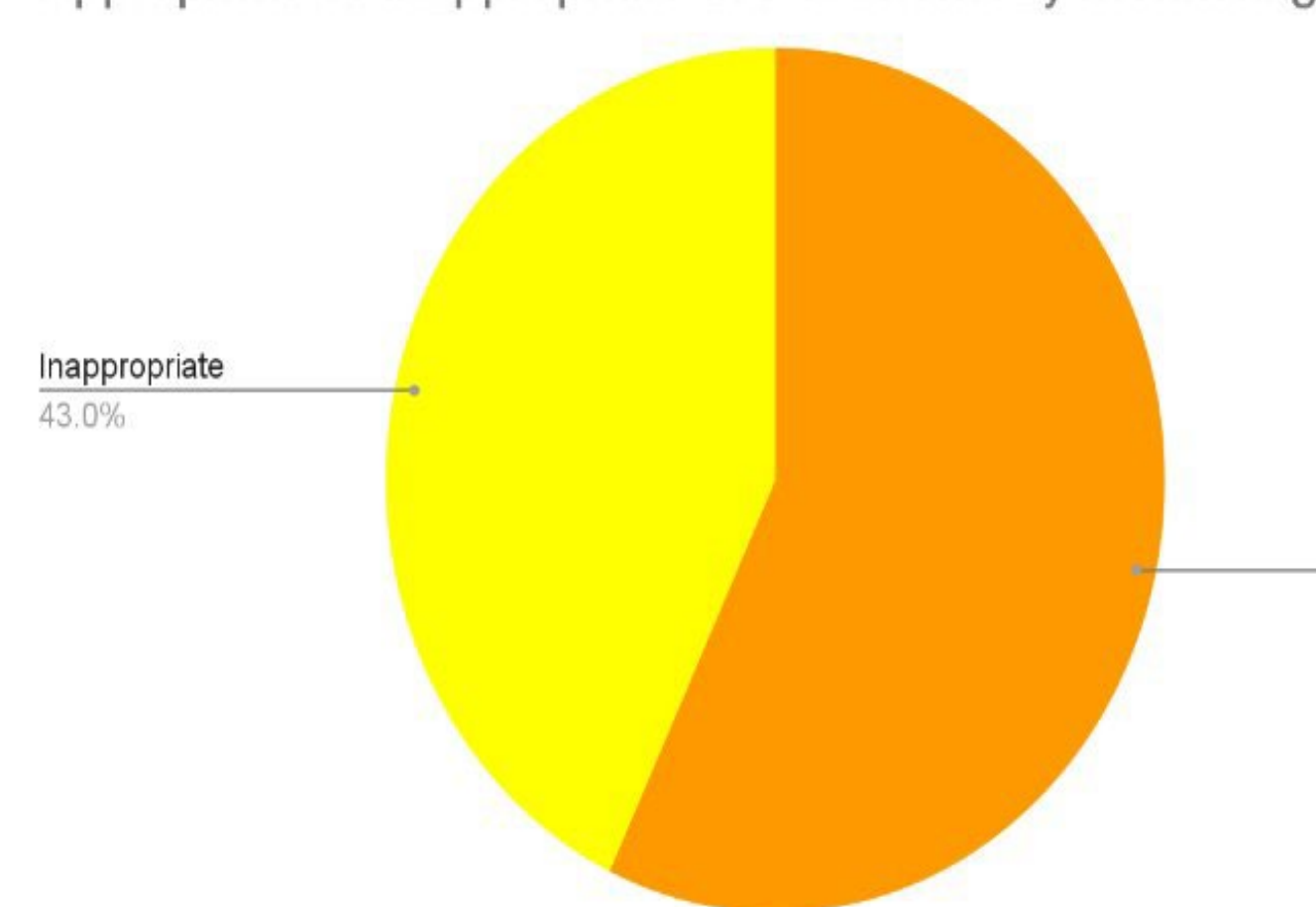
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Background

- This EBP aimed to evaluate a nursing telemetry protocol for discontinuing telemetry orders, to decrease inappropriate telemetry utilization in a neurology nursing unit.
- In the current climate of nurse staffing and supply shortages, nurses must lead the development of innovative ways to minimize waste and improve healthcare quality.
- The American Heart Association (AHA) published guidelines updated in 2017 that outline indications and the duration of appropriate telemetry usage.¹ The purpose of nursing protocols is to promote better patient outcomes, optimize patient safety, and improve the quality of care.
- Nurse-driven protocols assist nurses in integrating evidence-based practices into patient settings, support nurses' decision-making capacity, and increase their autonomy and accountability.²

Appropriate vs Inappropriate use of telemetry monitoring



Graph 1 Appropriate vs Inappropriate use of telemetry

Data from Henry Ford Hospital Providence- Novi Campus

November 15 - December 15, 2023

- Average of **105** monitored patients/day; with \$1400/day cost
- Mean average hours of telemetry usage **100.825** hours during pre-implementation phase.
- If 43% are inappropriate = **\$63,210/day waste**
- \$23 million/year waste**
- Multiply that waste per market = multi-millions!

Purpose/Aim

To ensure safe monitoring of patients on telemetry and provide guidelines for continuing and discontinuing telemetry monitoring.

Nurse-Driven Protocol

- The physician orders Telemetry monitoring and monitors following the guidelines outlined in the Telemetry Monitoring policy.
- Patients will be evaluated each shift for the necessity of telemetry monitoring based on specific criteria listed below by the registered nurse (RN).
- Outside of the ICU, PCU, and ER, telemetry monitoring can be discontinued without an additional MD order after the patient meets the criteria set within the protocol.

Methodology

Initiating a 4-part iterative EBP process improvement using the Plan-Do-Study-Act (PDSA) model offers a method of testing to achieve improvements on a small scale.

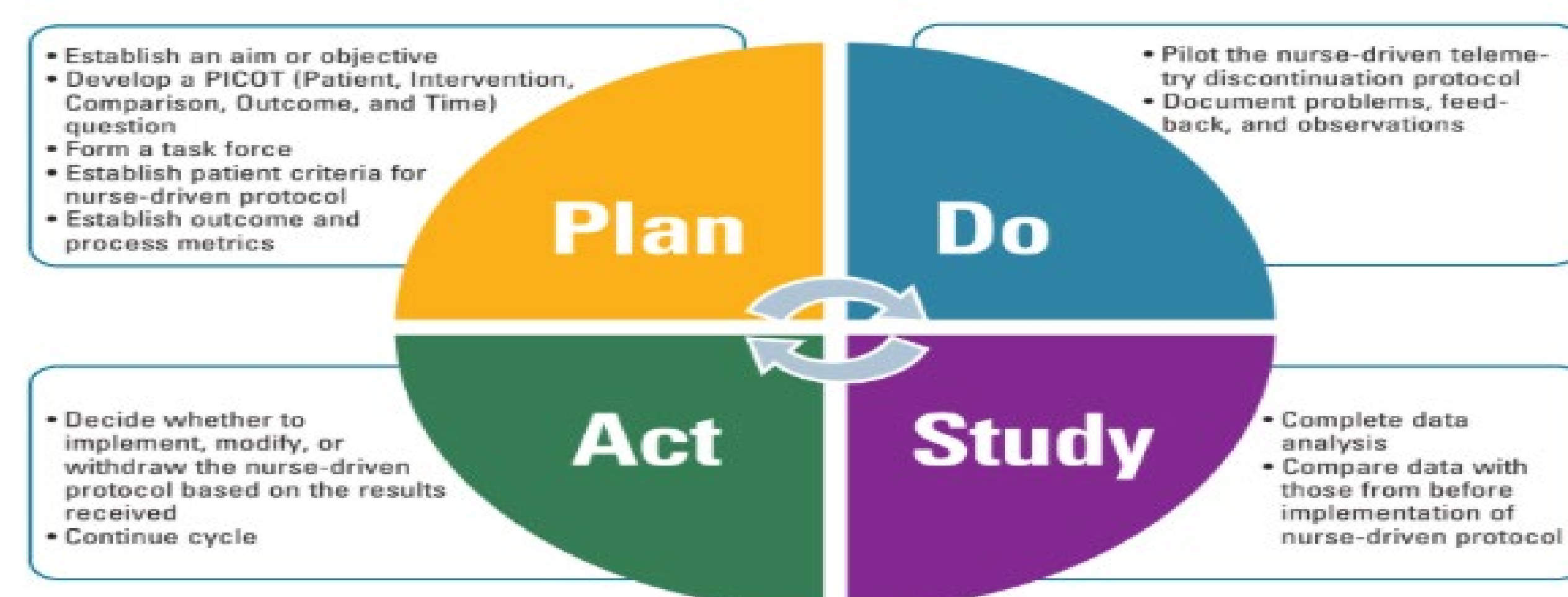
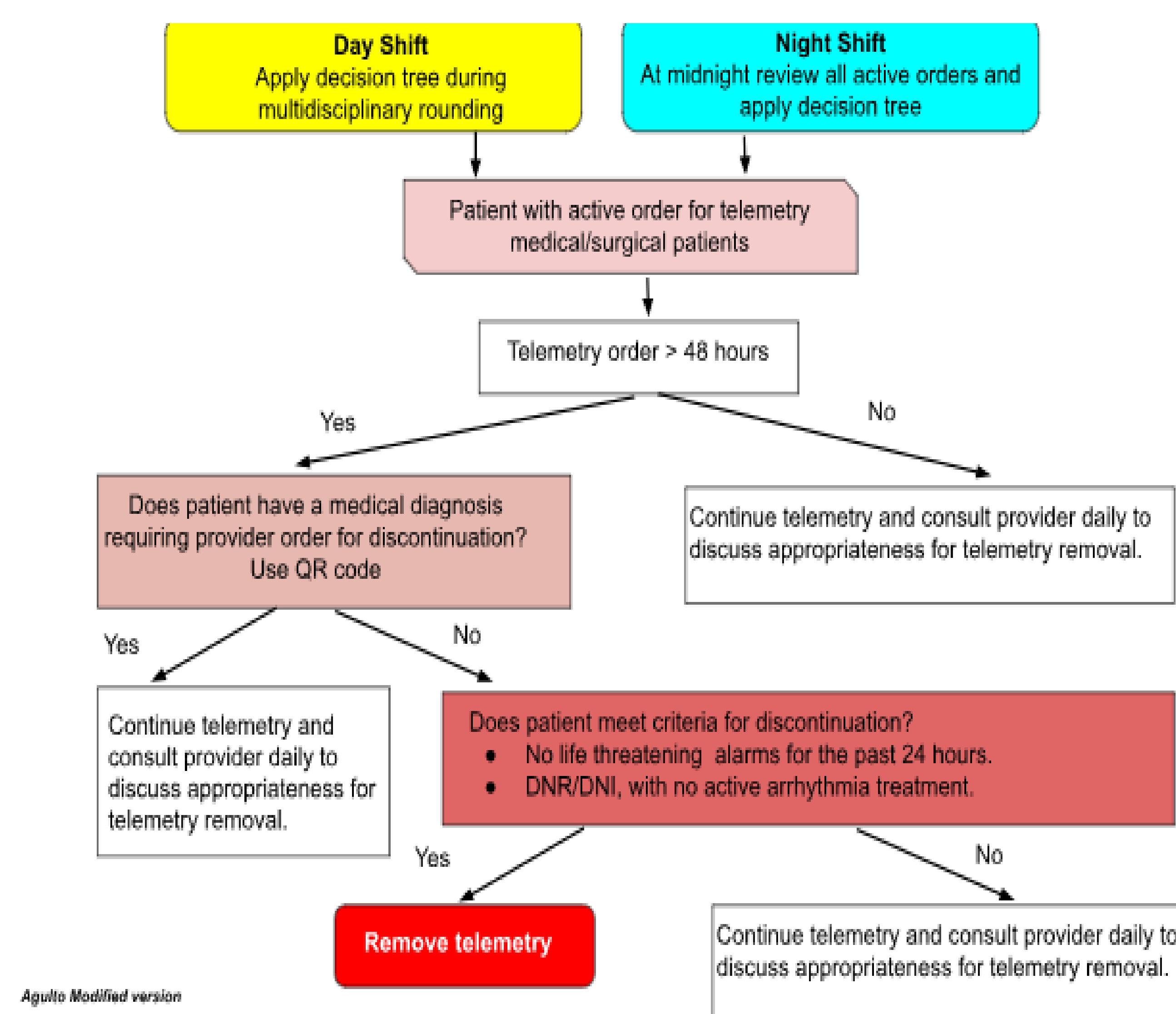


Figure 1 The Plan-Do-Study-Act Cycle

Methodology (continued)

- Current data revealed that from March to May, Ascension Novi Campus had an average usage of 105 telemetry boxes with average utilization hours of ≥ 96 hours of telemetry. There was no documented data of any delay in admission from emergency. Still, there were documented events when SWAT nurses were asked to round on units and triage patients that can be discontinued from telemetry in March 2023 to January 2024.
- The proposed project will only be implemented on 5A and 5B neurology, and medical-surgical units, with a total of 32 beds. IRB approval will be obtained before the start of the project. As the process owner, the author will gather two weeks of pre-implementation data about the telemetry usage of each patient in these two units, which includes the patient's DRG and length of telemetry usage. While the pre-implementation audit is going, an educational initiative will be shared for nurses and physicians about the two-week trial of the Nurse Driven Protocol telemetry discontinuation.



Agulto Modified version

Figure 2 Nurse Driven Telemetry Discontinuation Decision Tree

Results/Outcome

The primary outcomes will be the duration of telemetry monitoring during the pre- and post-intervention periods. Secondary outcomes were the hospital Length of Stay (LOS) projected cost avoidance of telemetry Monitoring between the two study periods (pre- and post-intervention).

Pre-Implementation:

December 2023

Telemetry mean average hours of
usage• **100.825 (including outliers)**

Average LOS

• **4.719**

Implementation:

January-February 2024

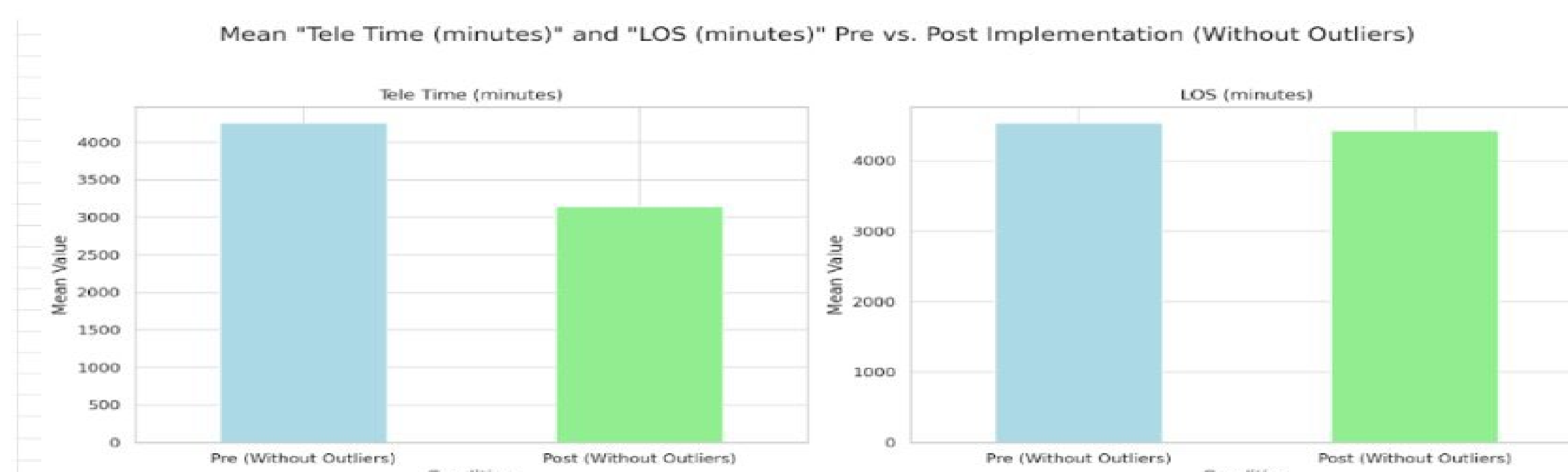
Telemetry mean average hours of
usage• **59.41 (including outliers)**

Average LOS

• **4.278**

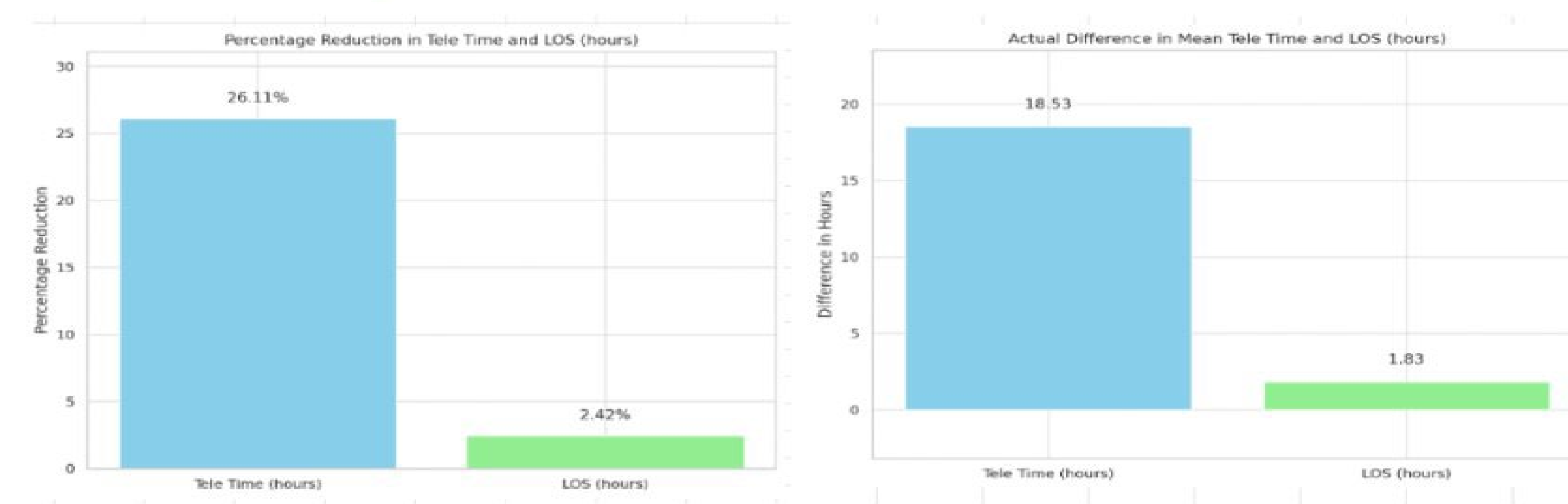
Results/Outcome

Comparative Analysis



Graph 1 Comparative Analysis Pre and Post Implementation Without Outliers

Statistical Analysis Result



Graph 2 Statistical Analysis Result

Conclusions/Implications for Practice

Improving quality patient care through the development and implementation of clinical nursing protocols is supported by numerous studies as an essential component of health care that promotes the efficient use of resources.

Reducing Institutional Financial Burden:

- A 0.432-hour shorter LOS in the implementation phase results in cost savings of **\$1,245 per patient with outliers**.
- A **1.83** shorter LOS in the implementation phase results in cost savings of **\$ 5,273.96 per patient without outliers**.
- During the implementation phase, telemetry usage was reduced by 41.41 hours (1.73 days), saving us **\$2415.58 per patient with outliers**.
- Telemetry usage was reduced by 18.43 hours (0.76 days), saving us **\$1061.18 per patient without outliers**.

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